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RESEARCH ARTICLE

## A Deep Learning-based System for Detecting Anemia from Eye Conjunctiva Images taken from a Smartphone

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### ABSTRACT

Anemia is a severe health condition commonly prevalent among women of reproductive age and children below five years. Screening patients before the condition becomes critical and can save many lives. World Health Organization (WHO) has set the “Global nutrition target 2025-anemia,” aiming to reduce 50% of anemia cases among women of reproductive age. This target can be achieved through a time-efficient, cost-effective, and easy-to-use tool. Traditional testing methods require specific chemicals, machines, and equipment that are not available everywhere. It also requires the presence of nurses, laboratory workers, and doctors. These methods are costly, time-consuming, and produce biohazard waste, thus polluting the environment. We developed an Artificial Intelligence (AI)-based bot that can be used for screening people for anemia. The bot service is based on two models: a segmentation model to segment the Region of Interest (ROI) and a classification model to classify anemic cases from normal ones. To train the model, we have collected data from 160 anemic and 140 non-anemic persons. In this paper, we have explained the architecture of the models, all the training parameters, and their deployment on cloud services using the REAN chatbot service. We manage to reach an Intersection Over Union (IOU) score of 0.922 for the segmentation model; validation recall of 0.95 and validation accuracy of 0.9699 for the classification model. This system is easy to use and does not depend on the availability of comprehensive laboratory infrastructure or trained personnel and thus can enable screening of anemia in low-resource settings.

### KEYWORDS

Anemia; Chatbot; Convolution neural networks; Deep learning; Image classification; Image segmentation; Non-invasive disease diagnosis; Transfer learning; U-Net

## 1. INTRODUCTION

Body ANEMIA is a condition in which the number or size of red blood cells, or the hemoglobin (Hb) concentration, falls below an established cut-off value, consequently impairing the capacity of the blood to transport oxygen around the body. The limited supply of oxygen in the body reduces physical and mental capacity. It raises the rates of maternal mortality and miscarriages in pregnant women and increases the occurrence of stillbirths, infant mortality, and low birth weight [1]. The Hb level needed to meet the physiological needs is differentiated according to gender, age group, and pregnancy condition. The major causes of anemia are nutritional deficiencies (particularly iron, folate, vitamins A, and B12 deficiency), hemoglobinopathies, and infectious diseases (such as malaria, tuberculosis, HIV, and parasitic infections) [2]. It is a severe condition worldwide; about 40% of pregnant women and 39.9% of children worldwide are anemic. Conditions are worse in South Asia, Central Africa, and West Africa. In India, 53% of women of reproductive age and 54% of children between 6 months to 4.9 years are anemic [3]. WHO targets to reduce anemia by 50% among women of reproductive age as per

their “Global Nutrition Target 2025- Anemia” [4]. To achieve the given target, there is a requirement for time-efficient, cost-effective and easy-to-use tools to perform mass screening. However, the current standard methods for hemoglobin testing like Sahil’s test, Hemoglobin-cyanide Method, Vanzeetti’s Azide Methemoglobin, and Reagent-less method do not provide compelling evidence to achieve the same. Moreover, the chemicals employed in the tests are biohazardous and pollute the environment [5]. In India, the hemoglobin test costs up to Rs.100 as of 12 July 2021 [6].

Doctors also perform four levels of physical examination to confirm anemia in a patient. These include a colour tint of the lower eyelid conjunctiva, nail-bed rubor, nail-bed blanching, and palmar crease rubor. The colour tint of these regions does not depend on the presence or absence of melanocytes (melanin-producing cells) in the patient’s skin [7]. Conjunctival pallor presence can be directly correlated with the condition of anemia and the determination of hemoglobin level [8]. But physical examination done by doctors is subjective and requires years of experience.

In recent years, there has been increasing interest in developing machine learning and deep learning models for disease diagnosis, which are often more accurate and efficient than less experienced physicians [9]. In particular, several machine learning-based binary classification models have been developed for anemia detection from images of the eye conjunctiva. However, previous studies have faced several challenges that have limited their accuracy and applicability [10, 11].

For instance, Tamir *et al.* 2017 attempted to detect anemia from smartphone images of the eye conjunctiva by processing the red and green component spectra of the conjunctiva colour and comparing them against a threshold. However, their model achieved an accuracy of only 78.9%, which is insufficient for reliable medical diagnosis [10].

Similarly, Kasiviswanathan *et al.* [12] used a digital image of the lower palpebral conjunctiva and basic details like age, sex, height, weight, and BMI to predict hemoglobin levels. Although they achieved a Pearson coefficient of 0.705, the measurement of height and weight may not be possible in home settings, and the dataset of 135 participants was small and unbalanced, leading to overfitting [12].

Mannino *et al.* developed an app for detecting anemia using finger nails with 265 participants, achieving an accuracy of  $\pm 2.4$  g dL<sup>-1</sup> and a sensitivity of 97%. However, the colour of the nails can change due to various factors, making this method unreliable. These challenges highlight the need for further research to develop more accurate and accessible machine learning-based methods for anemia detection. Also, the use of a specific device attached to a phone application is limited by device availability, and manual segmentation is not user-friendly [13].

There is an imbalance in the distribution of healthcare resources in society. Even after so many advancements, people still find it difficult to avail themselves of timely healthcare services because of geographical, time and financial constraints. In recent times, there has been a rise in digital infrastructure in India (2022), with a high penetration level of smartphones at 66%, and of social media platforms, for example, WhatsApp, at 690 million users [14]. Therefore, taking advantage of this digital infrastructure, healthcare-based chatbots can play a very helpful role in providing medical consultancy and disease diagnosis services in low-resource areas [15]. Such devices are also helpful in self-diagnosis and disease management [16]. Telegram and WhatsApp services are easy

to use and require minimal skills and resources from frontline workers. Hence, we have developed a chatbot service on Telegram and WhatsApp platforms by deploying a segmentation and classification model on the cloud systems and using the REAN Health architecture [17]. The segmentation model segments out the ROI, *i.e.* eye conjunctiva, from the image given by the user. The segmented image is fed into the classification model, which has convolution layers and fully connected layers. Convolution layers in the model extract features from ROI and fully connected layer classifies extracted features as anemic or non-anemic. The rest of the paper is organized as follows. Section 2(a) contains the complete information about the data collection, pre-processing, labelling and augmentation. Section 2(b) is about the architecture of the segmentation model, the discussion of training parameters, and setting up the output pipeline from the model to get the final cropped ROI. Section 2(c) is about the architecture and training of the classification model. Section 2(d) briefly describes the deployment of models. Section 2(e) states the training requirements of the models. Section 3 explains the various evaluation parameters used and the final results of the models. Section 4 discusses the use cases, future perspectives, and the conclusion.

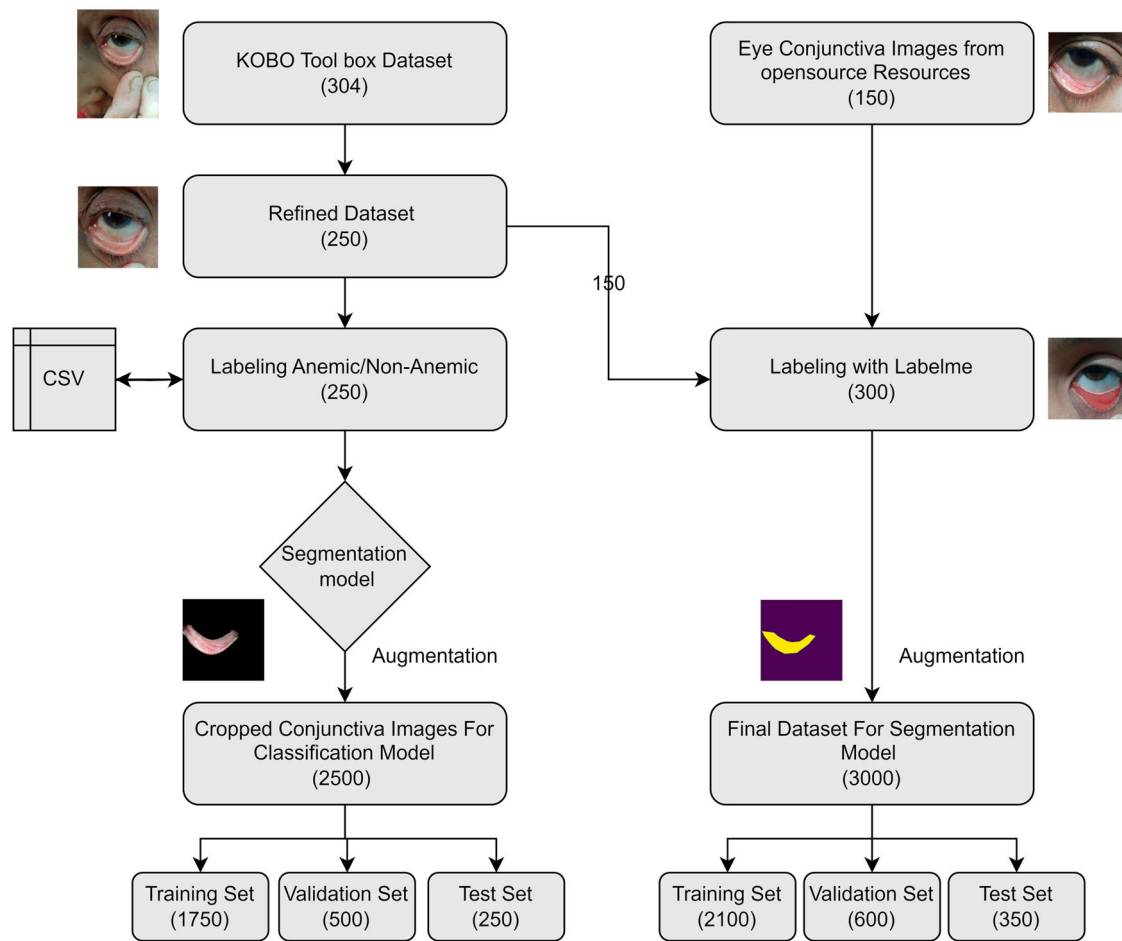
## 2. MATERIAL AND METHODS

### 2.1 Data Preparation

Data preparation is the most essential and critical step for making AI models work accurately. Data preparation includes data collection, quality checking, data labelling, and structuring into training, testing and validation sets. Data should be balanced among all the classes and free of any biases, such as regional bias, age group bias, gender bias, technology bias, and subjective bias. The complete strategy and steps are described in Figure 1 and the following sections.

#### 2.1.1 Data Collection

We used the Kobotoolbox for the data collection from the field. The Kobocollect app can be used to collect data without the internet and send the collected data later to the project when the internet is available [18]. We collected the images of left and right eye conjunctiva and the information of gender, age, and the Hb value in the anemia data collection project. There were 160 anemic cases and 140 non-anemic cases of different ages. One hundred six of the anemic cases were female, and the remaining 54 were male; among the non-anemic cases, 121 were male, and 19 were female. Detail about the distribution of the dataset is given in Table 1. We also



**Figure 1:** Complete workflow of creation of dataset for bot segmentation and classification model. The value inside the bracket indicates value of data division

used some open-sourced eyes images with conjunctiva parts available on the internet, specifically for segmentation models. The study was ethically approved by the IEC of Dayanand Medical College and Hospital Ludhiana (Reference No: IEC 2021–655 Dated: 31-May-2021). We consider people of mixed age groups, gender and ethnicity, and eliminate people with an eye infection, extreme drinking and smoking habits and medical conditions. We asked people to sit under proper lighting and pull the lower eyelid from the middle with their index finger so the eye conjunctiva region could be adequately visible, as shown in Figure 2(a). We instruct people to look upward to avoid the disturbance of eyelashes. We avoided the shadow effect on the conjunctiva part as it may cause the miscalculation of the pixels.

### 2.1.2 Data Pre-processing and Labelling

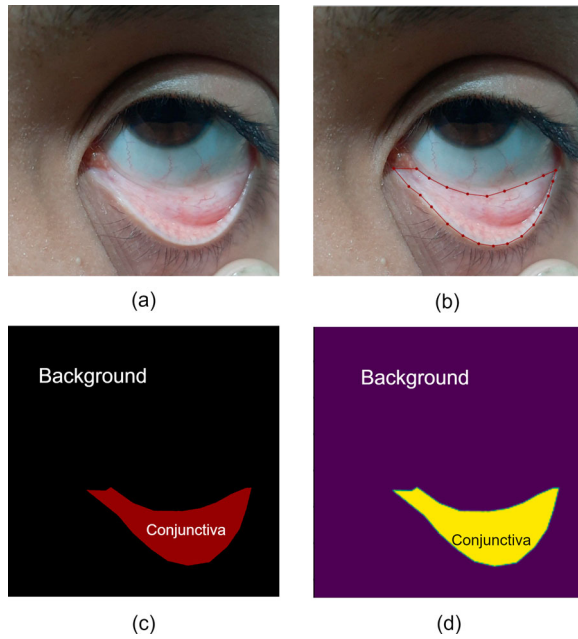
We reviewed each data point collected on the “Anemia data collection” project and eliminated the data points with blurred and half-shadow images. Images were center-cropped to eliminate unnecessary features and equalized for height and width. We randomly channelled

**Table 1: Data analysis**

| Age Groups | Anemic Non-Anemic |      |       |        |      |       |
|------------|-------------------|------|-------|--------|------|-------|
|            | Female            | Male | Total | Female | Male | Total |
| 0 - 19     | 9                 | 1    | 10    | 16     | 28   | 44    |
| 20 - 39    | 76                | 12   | 88    | 29     | 46   | 75    |
| 40 - 59    | 30                | 4    | 34    | 5      | 22   | 27    |
| > 60       | 6                 | 2    | 8     | 4      | 10   | 14    |
| Total      | 106               | 54   | 160   | 19     | 121  | 140   |

a few photos into the segmentation model dataset folder for labelling.

For the segmentation model: We developed ground truth using the software annotation tool “Labelme” [19] It can be used online and offline on a local computer to annotate pictures. Ground truth can be marked by making polygons around the ROI with a mouse click shown in Figure 2(b) Labelme returns a JSON file format. Ground truth can be reconstructed from the JSON file using the Python command “labelme json to dataset” as a batch command. The obtained ground truth is in an RGB image format as shown in Figure 2(c), but model training was



**Figure 2:** Detailed information about the process of labelling the ROI (a) the image given by user (b) creation of polygon on ROI (c) the output ground truth (d) grey scale ground truth to give as input to the model

done by converting it to a grayscale image with a value 0 for “background” and 1 for “conjunctiva” as shown in Figure 2(d). For classification model: Images were passed from the developed segmentation model to get the ROI, *i.e.* eye conjunctiva, and labelled as 0 for anemic and 1 for non-anemic according to its respective HB level, gender and age group.

### 2.1.3 Data Augmentation

Convolution neural networks require large datasets for training. Collecting and training medical datasets are expensive and time-consuming. Data augmentation is a viable solution for increasing the quantity and quality of the dataset. Data augmentation was done to avoid the overfitting of a model by teaching the invariance and robustness properties. There are various algorithms available for augmentation, mainly of two types: position-based algorithms and pixel-based algorithms. Majorly position-based algorithms (for example, rotation from 30 degrees to 60 degrees, horizontal and vertical flipping, a transitional shift in the x and y direction, zoom out and zoom in) were used. Two pixels-based algorithms: Gaussian noise and blurring, were used to consider the camera quality of users [20].

### 2.1.4 Data Splitting

The dataset was split in the ratio of 70:20:10 for training, validation and testing sets, respectively. Random selection has been ensured by using the test train split

function of the SKlearn library for splitting the data to avoid human biases.

## 2.2 Segmentation Model

### 2.2.1 Model Architecture

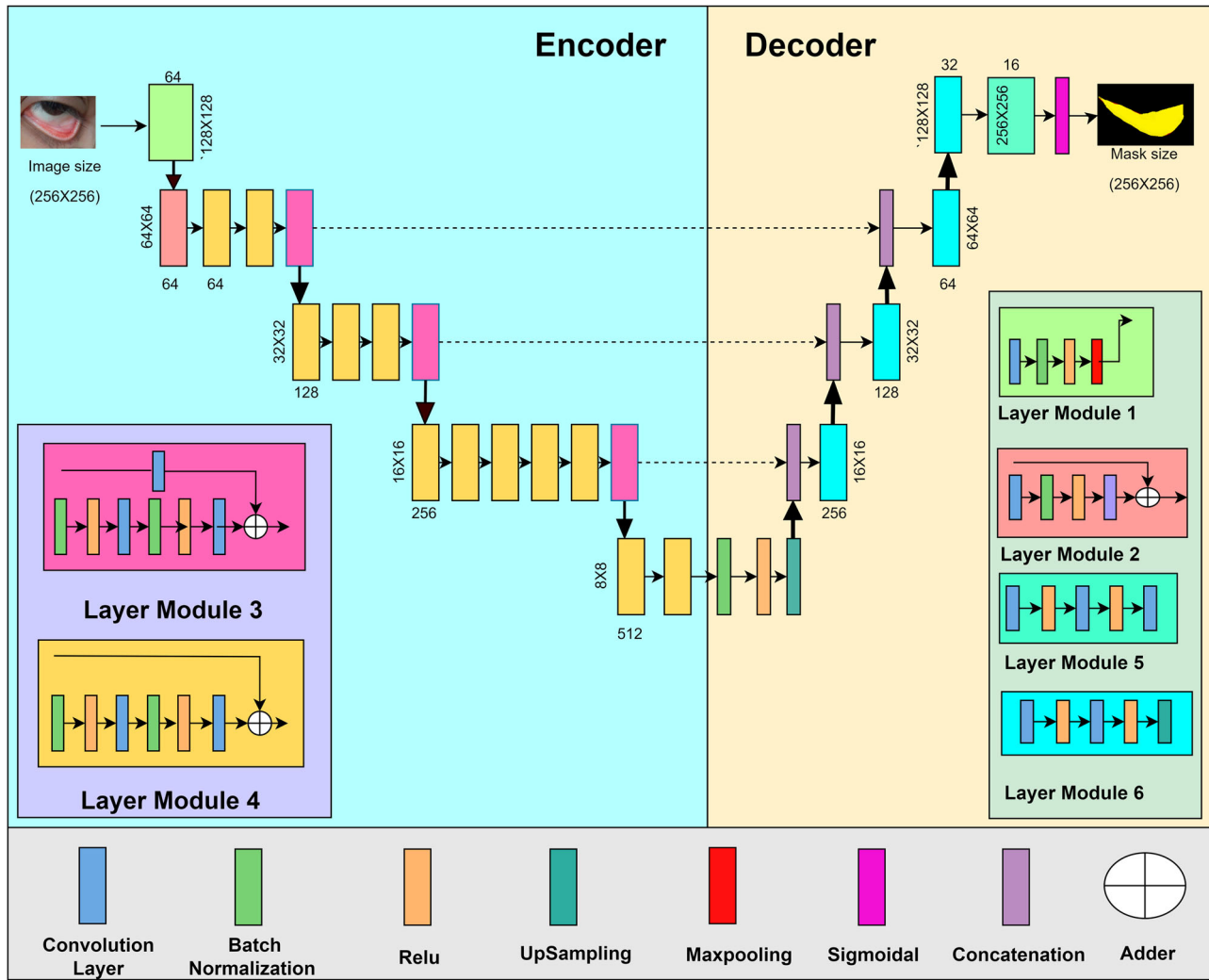
Recently, fully convolutional networks (FCN), with VGG16, AlexNet and Resnet and other neural networks pre-trained on Imagenet or Micronet weights as a backbone, have improved to give better results [21] and are used for segmentation in various biomedical applications [22]. We also used FCN, which has the contracting part (encoder) and expansive part (decoder); both are symmetrical to one another and form a U-shaped architect inspired by U-Net [23], as shown in Figure 3. Encoder extracts the spatial feature of conjunctiva from the image, and the decoder constructs the segmentation map from the extracted spatial features.

We built the encoder on the left side with Resnet34 backbone pre-trained on Imagenet weights. It consists of a sequence of blocks for downsampling. Each block is a combination of “Layer Module 1”, “Layer Module 2,” and “Layer Module 3”. These modules are made with a combination of  $3 \times 3$  convolution layers,  $2 \times 2$  max-pooling layers with a stride of 2, activation function Rectified Linear Unit (ReLU) and batch normalization layers with residual connections, as shown in Figure 3. The skip connections in ResNet with residual learning increase the depth and width of the network.

The expansive part (decoder) on the right side has four blocks; each block has an upsampling layer of the feature map, followed by a  $2 \times 2$  convolution (“up-convolution”) that halves the number of feature channels, a concatenation with the correspondingly cropped feature map from the contracting path via skip connection represented with dotted arrows, and two  $3 \times 3$  convolutions, each followed by an Activation function ReLU, as shown in Layer Module 5 and Layer Module 6. The final step was built with  $1 \times 1$  convolution and activation function sigmoidal to map 16 component feature vectors to the final output.

Skip connections are for recovering the spatial resolution in Deep layers by concatenating the low-level features of the contracting path with the high-level features of the expansion path. High-resolution features were combined with the upsampled output, causing a more precise feature assembly. The appropriate combination of long and short connections helps the network extract features at different levels and enhances the ability of expression while complementing semantic information at the high level and refining segmentation outlines at the low





**Figure 3:** Representation of the architecture of the segmentation model

level [24]. It was explicitly designed for Biomedical Image segmentation. It can precisely segment the ROI despite training on a minimal data set in less time. Dice loss was considered for training. The model was pre-trained on weights from “Imagenet”.

### 2.2.2 Output Pipeline

When the input image of size 256 \* 256 is sent to the model, it will return the predictive mask, as shown in Figure 4. The mask will have a pixel value ranging from 0 to 1. We plotted a histogram plot for the predictive mask and found the background pixels accumulated between 0 and 0.003 and pixels for conjunctiva distributed between 0.004 and 1. Pixels belonging to the background were set to 0, and pixels belonging to the conjunctiva were set to 1. When the pixel of the image ( $pi$ ) is multiplied by the refined mask’s corresponding pixels:

$$pi * 0 = 0 \quad (1)$$

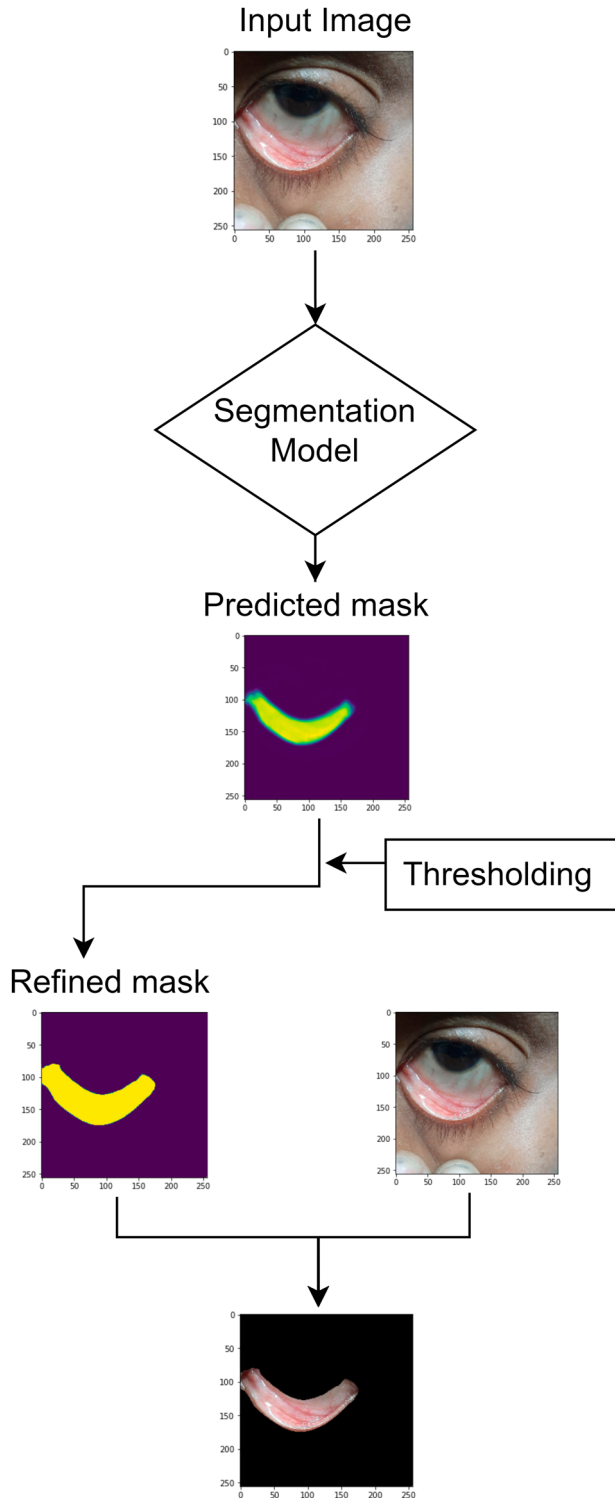
$$pi * 1 = pi \quad (2)$$

All the background pixels will be set to 0, and we will obtain the image with the ROI of size 256 \* 256.

### 2.2.3 Model Architecture

We built a convolution neural network (CNN) for performing the classification task. It comprises multiple non-linear layers explained below.

- **Input Layer:** It accepts the raw image and forwards it to the downstream layers for extracting features.
- **Convolution layer:** The convolution layers represent the input signal  $X$  into multiple feature maps by convoluting the input with multiple learnable filters. These multiple feature maps then flatten into a single vector to feed into Classifiers.



**Figure 4:** Flowchart of the complete process of getting cropped images with ROI

- ReLU : (Rectified-Linear Unit) is a piecewise linear function. This layer will output the input directly if it is positive. Otherwise, it will output zero.
- Dropout: The dropout layer randomly sets some input units to 0 with a frequency of rate decided at each step during training time; this helps prevent the overfitting

of models. Inputs that are not set to 0 are scaled up by  $1/(1 - \text{rate})$  so that the sum over all inputs remains unchanged.

- Batch normalization: the Batch normalization layer is used to standardize the inputs to a network, applied to either the activations of a prior layer of inputs directly. It accelerates training and provides some regularisation by reducing generalization error.
- Global average Pooling: The 2D Global average pooling block takes a tensor of size  $(W * H * D)$ ,  $W$  is image width,  $H$  is image height, and  $D$  is image depth. It computes the average value of all values across the entire  $(W * H)$  matrix for each of the  $(D)$ .
- Fully Connected Layer: In these layers, all the inputs from one layer are connected to every activation unit of the next layer. The final layer is fully connected; it takes up high-level filtered images and applies weights, thus translating them into labels with categories.
- Softmax layer: This layer is just before the output layer. This layer gives the decimal probabilities for each class. Those decimal probabilities are between 0 and 1.

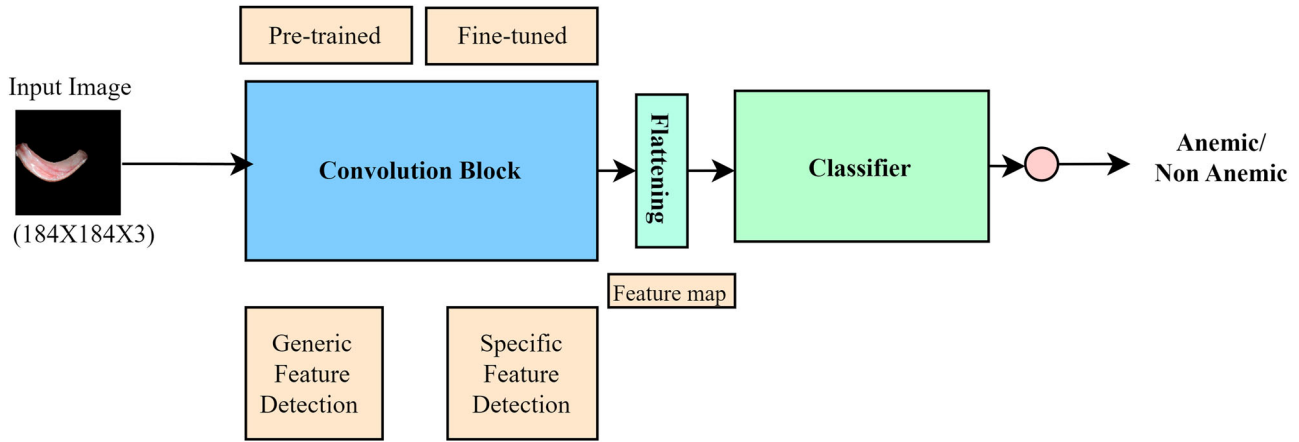
The first four layers mentioned above are used for feature extraction and others for classification [25] as shown in Figure 5. Binary cross-entropy is used as a loss function and Adam as Optimiser. It compares the predicted probability with the actual value, either 0 or 1, representing each class and penalises the model to get an accurate value. As it contains the log functions, the penalty is more for higher differences and lower for lesser differences.

### 2.2.4 Transfer Learning

Recently, transfer learning technology has been extensively utilized for medical imaging analysis. One commonly used technique is to pre-train a neural network on the source domain and then fine-tune it based on the instances from the target domain. Transfer learning utilizes the knowledge implied in the source domain to improve the performance of the target domain. It works in the same way as learning piano is much easier for a person who is already well versed in playing the violin compared to an amateur because the violin and the piano both are musical instruments and hence share many things in common [26]. The most commonly used source domain is ImageNet, an image database containing more than fourteen million annotated images with more than 20,000 categories.

## 2.3 Deployment of Models and Integration with the REAN Bot

We created an anemia API using Python's Fast API, custom Docker Image and Amazon's ECS Fargate service.



**Figure 5:** Representation of the architecture of the classification model

We kept the models on AWS cloud services [27]. Users can provide the image of eye conjunctiva on the REAN HealthGuru chatbot service on Telegram or WhatsApp. It will be processed by the REAN frontend wrapper and dialogue flow, and the required information will be sent to anemia result API to obtain the results and send them back to the user, as shown in Figure 6.

## 2.4 Model Training

The Model training was carried out on Google Colab Pro. The codes were mainly implemented in Python, including data pre-processing and algorithm implementation. The main libraries used were NumPy, Pandas, and OS for data handling and pre-processing; CV2 for image processing; TensorFlow for Model training and optimization; and Matplotlib for visualization.

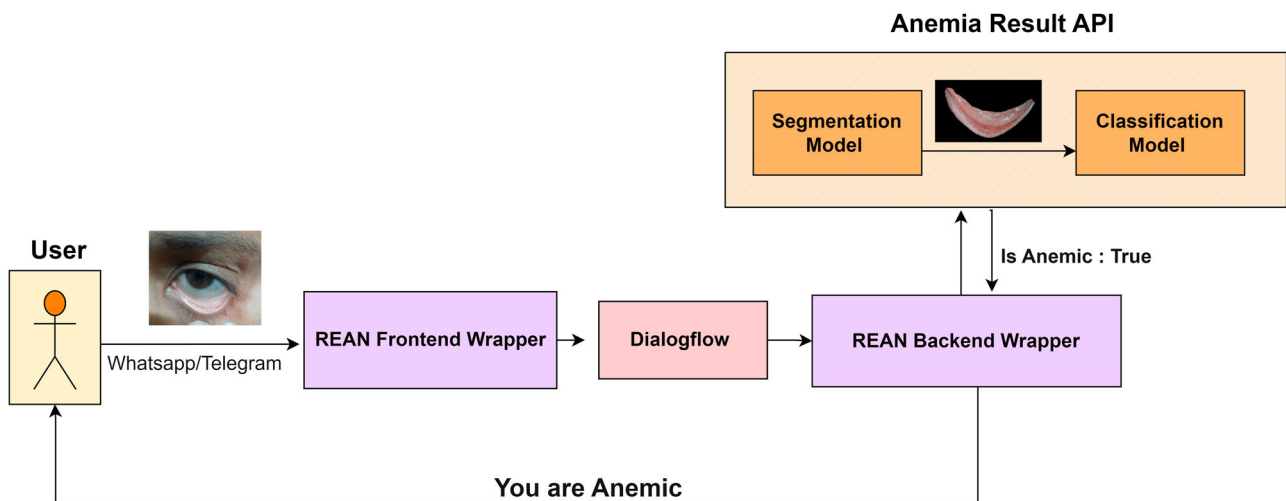
## 3. RESULTS AND DISCUSSION

### 3.1 Segmentation Model

We used Intersection Over Union (IOU), Mean Intersection Over Union (MIOU) and F1 score as a validation metric to measure the accuracy of the proposed segmentation model. These validation metrics are widely used in computer vision, particularly for object detection challenges. The IOU value is calculated between the ground truth and the predicted mask.

$$\text{IOU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (3)$$

The model's training experimented with VGG16, ResNet-50, ResNet-34, and ResNet-18 as backbones with fixed batch size, epochs and learning rate; detailed results are



**Figure 6:** Flow chart from user to provide image to get the final result



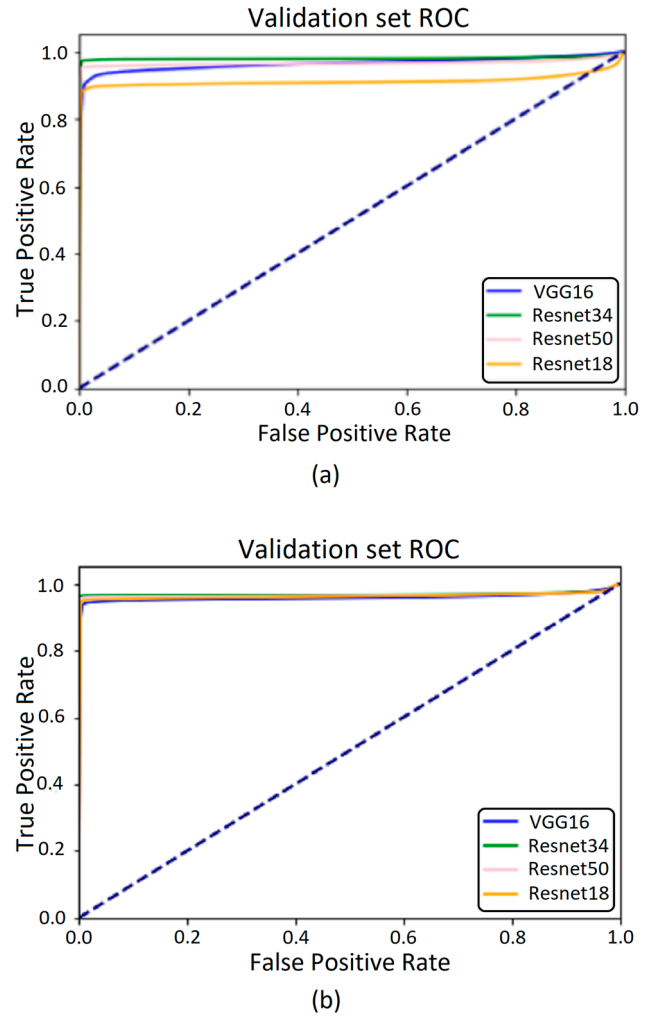
**Table 2: Selecting the architecture for segmentation model**

| Model    | Augmentation | AUC    | IOU    | MIOU   | F1 score | Parameters |
|----------|--------------|--------|--------|--------|----------|------------|
| VGG      | yes          | 0.9578 | 0.8659 | 0.6959 | 0.9280   | 23,748,241 |
|          | no           | 0.9636 | 0.7726 | 0.6763 | 0.8706   | 23,748,241 |
| Resnet34 | yes          | 0.9670 | 0.9272 | 0.9099 | 0.9621   | 24,438,804 |
|          | no           | 0.9585 | 0.9057 | 0.8945 | 0.9501   | 24,438,804 |
| Resnet50 | yes          | 0.9642 | 0.9079 | 0.9074 | 0.9515   | 32,513,556 |
|          | no           | 0.9648 | 0.8907 | 0.8927 | 0.9418   | 32,513,556 |
| Resnet18 | yes          | 0.9615 | 0.8949 | 0.9095 | 0.8196   | 14,330,644 |
|          | no           | 0.9113 | 0.8292 | 0.8868 | 0.9064   | 14,330,644 |

presented in Table 2. ResNet-34 has shown the best performance among those models. It has an Area Under Curve (AUC) of 0.9670, IOU of 0.9272 and MIOU of 0.9099. The second-best model, ResNet50, with an AUC of 0.9642, IOU of 0.9079 and MIOU of 0.9074. It also has comparatively fewer trainable parameters, 24,438,804, compared to ResNet50, which has 32,513,556 trainable parameters. The models have been trained with and without augmentation; training with an augmented dataset had shown better results than can be justified with the better F1 score of 0.9621 when augmentation was allowed as compared to 0.9501 when trained with an unaugmented dataset could be seen from the Receiver Operating Characteristic (ROC) curves in Figure 7.

After selection, the model training was executed on different batch sizes (4, 8, 16, 32, and 64). While smaller batch sizes 4 and 8 increased the training time to 18s and 14s per epochs, respectively. However, performance was more or less the same. Batch size 16 showed comparable performance with IOU 0.9214 less execution time of 10s. Although Batch size 32 has shown the best IOU of 0.9273 and less training time of 9s per epoch. but it requires high RAM and more number of epochs for training. The proposed model was then trained with different learning rates. It was observed that a significantly lower learning rate did not reach accuracy even with a very high number of epochs, and a very high learning rate caused spikes in the learning curve. The optimal learning rate was selected at 0.001. The model requires 10ms to run an epoch and 150ms to run a single batch step.

A dual encoder-decoder based deep polyp segmentation network was used to segment colonoscopy images [28]. With their modified polyp dataset, the researchers obtained an MIOU of 0.897. Researchers have also used a network called Semantic Transformer Representation Network (STRNet) for crack segmentation, and another network called Attention-based Generative Adversarial Network (AGAN) was developed to precisely segment internal cracks in concrete at the pixel level [29, 30]. The



**Figure 7:** ROC curves of pre-trained models used as backbone of encoder part of UNet (a) training with unaugmented dataset (b) training with an augmented dataset

modified polyp dataset was used to achieve MIOU values of 0.846 and 0.93 with different parameters. Additionally, the Semantic Damage Detection Network (SDDNet) was developed specifically for the segmentation of cracks in concrete images, achieving an MIOU of 0.846 [31]. Our proposed network, Resnet34, obtained a higher MIOU of 0.9099, indicating that it performed better than previous studies. Similarly, the proposed network Resnet50 also achieved a higher MIOU of 0.9074.

### 3.2 Classification Model

For evaluating the performance of the classification model, we consider the ROC curve and three metrics: accuracy, recall and precision. Accuracy is the measure of how often a model is able to classify each class correctly.

It can be mathematically represented as follows:

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP} \quad (4)$$

TP represents the true positive cases (the number of the anemic cases is correctly identified as anemic cases), TN represents true negative cases (the number of non-anemic cases correctly classified as non-anemic), FN represents the false negatives cases (the number of anemic cases misclassified as non-anemic), and FP represents the false positives (the number of non-anemic cases misclassified as anemic).

The recall is the number of times in the given number of cases that our model is able to label anemic as “anemic”.

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

Precision measures how precisely the model categorizes the anemic as “anemic.” In other words, out of those predicted anemic cases, how many of them are actually anemic.

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

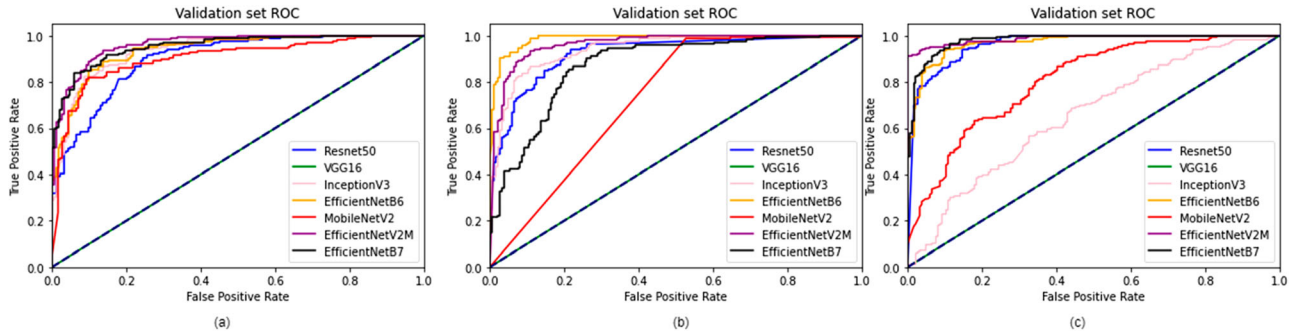
A ROC curve plots the fraction of anemic cases correctly classified as anemic and represented as a true positive rate (TPR) versus the fraction of non-anemic wrongly classified as the anemic is represented as a false positive rate (FPR). The closer the curve approaches the top left corner, the better the system performs. The most significant performance measure extracted from the ROC curve is the value of the area under the curve (AUC). The most optimal system has an AUC value of 1. We experimented with various pre-trained models for the feature extraction part to reduce the training time and computation cost. We experimented with three training scenarios.

### 3.2.1 Freezing all the Layers

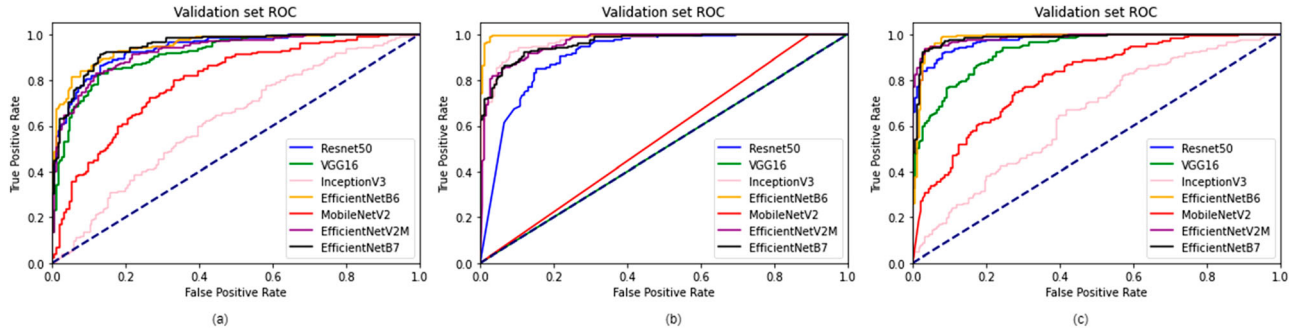
### 3.2.2 Training all the Layers

### 3.2.3 Freezing top layers (used for generalized features) and fine-tuning bottom layers (used for specific features)

At first, we used fully connected layers as a classifier and then Global average pooling; the ROC curves for all the scenarios are in Figures 8 and 9, and the detailed results are in Tables 3 and 4, respectively. Among all the scenarios, EfficientNetB6, EfficientNetV2M, and EfficientNetB7 have shown better performance when trained by freezing the top layers and fine-tuning the bottom layers with AUC of 0.970, 0.988, 0.98 and validation loss of 0.2218,



**Figure 8:** ROC curves for fully connected layers in different training condition (a) Freezing all layers (b) training all layers (c) Freezing top layers and fine tuning the bottom layers



**Figure 9:** ROC curves for Global average pooling in different training condition (a) Freezing all layers (b) training all layers (c) Freezing top layers and fine tuning the bottom layers

**Table 3: Global average pooling**

| Model           | Training           | AUC   | Loss   | Recal  |
|-----------------|--------------------|-------|--------|--------|
| ResNet50        | Non-Trainable      | 0.937 | 0.3197 | 0.8072 |
|                 | Trainable Freezing | 0.909 | 0.4668 | 0.8916 |
|                 | Top-Layers         | 0.976 | 0.3449 | 0.8855 |
| VGG16           | Non-Trainable      | 0.912 | 0.3814 | 0.8494 |
|                 | Trainable Freezing | 0.5   | 0.6975 | 1.0000 |
|                 | Top-Layers         | 0.931 | 0.3489 | 0.8735 |
| InceptionV3     | Non-Trainable      | 0.622 | 1.1040 | 0.3976 |
|                 | Trainable Freezing | 0.969 | 0.3926 | 0.8554 |
|                 | Top-Layers         | 0.648 | 0.7039 | 0.6024 |
| EfficientNetB6  | Non-Trainable      | 0.95  | 0.3166 | 0.8735 |
|                 | Trainable Freezing | 0.995 | 0.1234 | 0.9872 |
|                 | Top-Layers         | 0.983 | 0.1453 | 0.9337 |
| MobileNetV2     | Non-Trainable      | 0.792 | 0.5751 | 0.8072 |
|                 | Trainable Freezing | 0.552 | 8.5622 | 1.0000 |
|                 | Top-Layers         | 0.803 | 1.9344 | 0.7470 |
| EfficientNetV2M | Non-Trainable      | 0.925 | 0.3598 | 0.8373 |
|                 | Trainable Freezing | 0.965 | 0.1746 | 0.9217 |
|                 | Top-Layers         | 0.985 | 0.2405 | 0.9398 |
| EfficientNetB7  | Non-Trainable      | 0.947 | 0.3254 | 0.8916 |
|                 | Trainable Freezing | 0.964 | 0.1966 | 0.9819 |
|                 | Top-Layers         | 0.982 | 0.1518 | 0.9458 |

**Table 4: Different models with different training conditions using FCL as classifier**

| Model           | Training           | AUC   | Loss   | Recal  |
|-----------------|--------------------|-------|--------|--------|
| ResNet50        | Non-Trainable      | 0.9   | 0.4807 | 0.7590 |
|                 | Trainable Freezing | 0.8   | 0.3183 | 0.8675 |
|                 | Top-Layers         | 0.963 | 0.8892 | 0.9398 |
| VGG16           | Non-Trainable      | 0.5   | 0.6975 | 1.0000 |
|                 | Trainable Freezing | 0.5   | 0.6975 | 1.0000 |
|                 | Top-Layers         | 0.5   | 0.6975 | 1.0000 |
| InceptionV3     | Non-Trainable      | 0.935 | 0.3243 | 0.8133 |
|                 | Trainable Freezing | 0.942 | 0.3045 | 0.9036 |
|                 | Top-Layers         | 0.655 | 0.6473 | 0.7651 |
| EfficientNetB6  | Non-Trainable      | 0.937 | 0.4012 | 0.8554 |
|                 | Trainable Freezing | 0.967 | 0.2229 | 0.9458 |
|                 | Top-Layers         | 0.970 | 0.2218 | 0.8675 |
| MobileNetV2     | Non-Trainable      | 0.902 | 0.5083 | 0.7711 |
|                 | Trainable Freezing | 0.730 | 1.6307 | 0.8253 |
|                 | Top-Layers         | 0.807 | 1.9100 | 0.9157 |
| EfficientNetV2M | Non-Trainable      | 0.961 | 0.2733 | 0.9337 |
|                 | Trainable Freezing | 0.964 | 0.3448 | 0.9398 |
|                 | Top-Layers         | 0.988 | 0.1590 | 0.9458 |
| EfficientNetB7  | Non-Trainable      | 0.953 | 0.4225 | 0.9699 |
|                 | Trainable Freezing | 0.873 | 0.3871 | 0.8494 |
|                 | Top-Layers         | 0.98  | 0.1916 | 0.9458 |

0.1590 and 0.1916 respectively while using fully connected layers as classifier and AUC of 0.983, 0.985, 0.982 and validation loss of 0.1453, 0.2405 and 0.1518 respectively while using Global average pooling as a classifier.

Here, EfficientNetB6 and EfficientNetB7 has shown the least validation loss and comparable AUC value. Also, training parameters for EfficientNetB6 and EfficientNetB7 are 9,351,073, 11,536,961 and training time per epochs is 27s and 36s respectively. Hence, we concluded

that EfficientNetB6 is the best model for feature extraction with a global average pooling layer and fine-tuning the bottom layers as training conditions. Further, we fine-tuned the model for different batch sizes, no of epochs and learning rate to obtain minimum loss and maximum recall and accuracy.

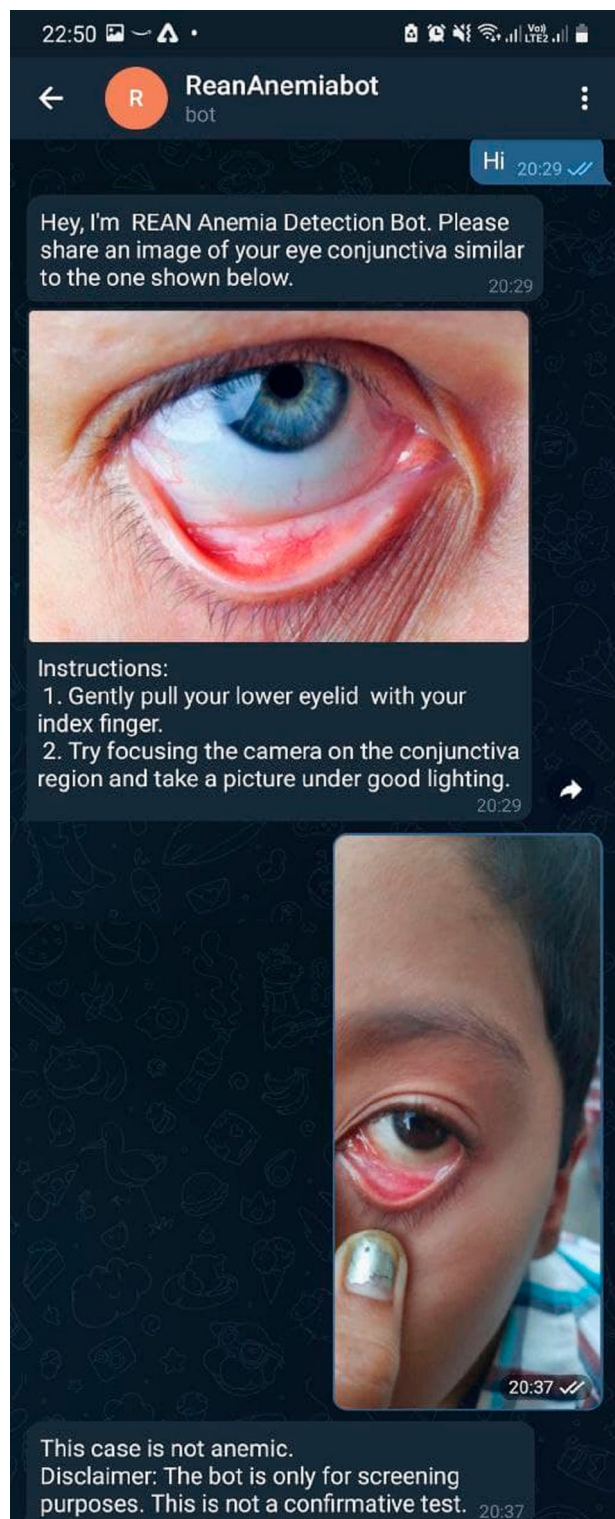
The final selected model has a training loss of 0.1376, training accuracy of 0.9625, training precision of 0.9651, training recall of 0.9639, validation loss of 0.1479, validation accuracy of 0.9699, validation precision of 0.9415, validation recall of 0.9571.

#### 4. CONCLUSION AND FUTURE PERSPECTIVE

The proposed AI-based chatbot gives results in only 35 s. It can be used by anyone, at any time, and for anyone. It can be used for mass screening to mark areas with severely anemic people. Governments can use it for making strategies, doctors for telemedicine, frontline health workers for reference, and the general public for self-diagnosis. Accredited social health activists (ASHAs) act as a critical link between the Healthcare delivery system and the community. They are acting as a worrier in child and maternal care. They facilitate mothers and neonates with the necessary facilities, maintenance, and knowledge. With the adoption of new technology, *e.g.* TeleMedicine, point-of-care diagnosis, and a surge of AI in the field, users can be helped with innovative and more effective solutions. Also, people in low-resource setups can benefit from better and quick outreach of assistance. Bot services are easy to use as compared to complex apps. The model's accuracy can depend on the lighting condition, the quality of the mobile phone camera, and the distance of taking an image. Some mobile phones use filters to smoothen photos. It can cause a complete miscalculation in calculating the pixel values. For training the model, we considered some reasonable assumptions. Therefore, we gave the exact instructions to the user for capturing the images, as shown in Figure 10.

In the future, the model can be improved to give better performance. Essential parameters such as gender and age can be considered while developing the model. The model can be upgraded to classify with the severity level and later to exact hemoglobin levels. We may require a more extensive dataset that is diverse among all age groups, genders and ethnicity. We may require a more extensive dataset diverse among all the age groups, both the gender and mixed ethnicity. We will welcome organizations to share the data points using the tool KoboToolBox with us, by taking the concerned person's consent.





**Figure 10:** Screen shot of working of the final chatbot service

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## DISCLOSURE STATEMENT

No potential conflict of interest was reported by the author(s).

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